

Scheduling five processes, part two

P1–P5 arrive one millisecond apart, starting at time 0. This works out the Gantt chart, turnaround time and waiting time under FCFS, non-preemptive SJF, and round robin with a quantum of 4.

Turnaround time = completion – arrival. Waiting time = turnaround – burst.

Process data

Process	Arrival	Burst
P1	0	6
P2	1	13
P3	2	7
P4	3	5
P5	4	15

P1 P2 P3 P4 P5

1. First come, first served



Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	6	6	6	0
P2	1	13	19	18	5
P3	2	7	26	24	17
P4	3	5	31	28	23
P5	4	15	46	42	27

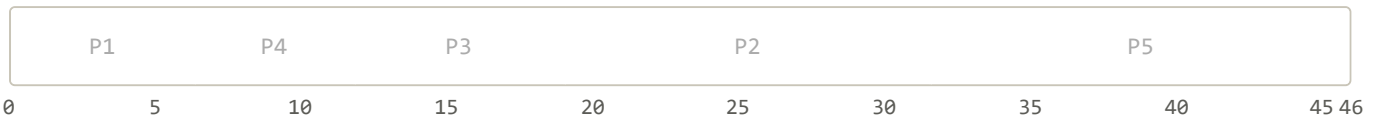
Average turnaround

Average waiting

23.6

14.4

2. Shortest job first



At time 6, P2 (13), P3 (7), P4 (5) and P5 (15) have all arrived, so the shortest remaining burst is picked each time a process finishes.

Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	6	6	6	0
P2	1	13	31	30	17
P3	2	7	18	16	9
P4	3	5	11	8	3
P5	4	15	46	42	27

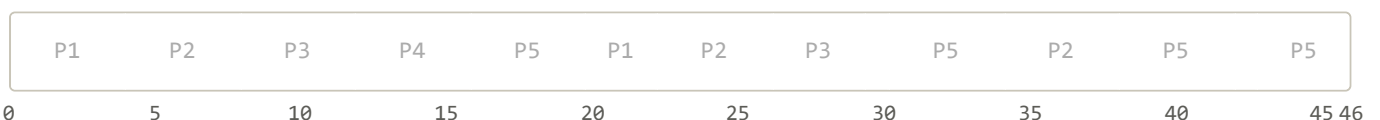
Average turnaround

20.4

Average waiting

11.2

3. Round robin (quantum = 4)



Process	Arrival	Burst	Completion	Turnaround	Waiting
P1	0	6	22	22	16
P2	1	13	43	42	29
P3	2	7	29	27	20
P4	3	5	30	27	22
P5	4	15	46	42	27

Average turnaround

32.0

Average waiting

22.8

Summary

Algorithm	Average turnaround	Average waiting
First come, first served	23.6	14.4
Shortest job first	20.4	11.2
Round robin (q=4)	32.0	22.8

SJF wins on average wait because it keeps clearing the shortest remaining job first, but P5 still waits 27ms under every policy here — it's the longest job and arrives last, so nothing shields it from that queue. Round robin costs the most on average because a 4ms slice against jobs this long forces several extra trips through the ready queue.